Forum:	World Bank (WB)
Issue:	Investing in resilient healthcare systems to mitigate future pandemic risks
Author:	George Kokkinakis
Position:	Deputy President

Personal Introduction

Dear Delegates,

Welcome to the 1st Moraitis School Model United Nations Conference! My name is George Kokkinakis, and I am an IB1 student at Moraitis School. I am deeply honoured to serve as a Deputy President of the World Bank at this year's MoMUN. MoMUN marks my second conference as a chair, and I am extremely grateful for this opportunity.

I, therefore, would like to commend each and every one of you for choosing to participate. Whether this is your first conference or your tenth, I will strive to ensure that everyone feels welcome to the beautiful world of MUN, and every voice is heard. This conference will encourage you to push yourself, testing your ability to debate, reason and negotiate on redesigning reality – abilities which will serve you well in any competitive environment.

This study guide offers essential insights regarding the investment in resilient healthcare systems to mitigate future pandemic risks, which will help you build a deeper understanding of the topic. That being said, I strongly encourage you to conduct independent research in order to fully grasp your own country's policy in relation to the topic.

Should any questions arise while preparing for the conference, please do not hesitate to reach out to me at kokkinakisgeorge10@gmail.com, as I am more than happy to assist you.

I look forward to meeting you all this October, and I wish you the absolute best conference.

Yours sincerely,

George Kokkinakis

Topic Introduction

Since 2020, when the COVID-19 pandemic started, the need for resilient healthcare systems has become more prominent than ever. Although many states have healthcare systems, as illustrated by the rapid construction of emergency facilities in China during the COVID-19 pandemic, these systems may still be incapable of handling a sudden surge in patients, due to a pandemic or a widespread illness. One estimate placed the annual cost of pandemic prevention and preparedness at $31.1 billion, while estimates of the COVID-19 pandemic's economic impact vary considerably, proving that investing proactively can be highly beneficial for the national healthcare provider of every state, and can lead to significantly fewer losses compared to those during a pandemic.

Pandemics have existed since the beginning of humanity, and have affected all the corners of the world. From the 1918 influenza virus, also known as the Spanish Flu, which infected an estimated one-third of the world's population and caused tens of millions of deaths, to the COVID-19 pandemic, which caused over 7 million reported deaths worldwide since 2019, and plague, which was pivotal in reshaping the societal and geopolitical context of its time, pandemics reshape the course of human history, and alter our societies. Besides the sociocultural impact of pandemics, these widespread diseases affect the economy of each and every state infected. According to Statista, a global data and business intelligence platform, during 2020, the global Gross Domestic Product (GDP) contracted by approximately 3%, meaning that more than 2 trillion United States Dollars of economic output were lost. Consequently, it is visible that the social and economic impact of pandemics is very broad, and it can derail a state's development.

A way in which said consequences that stem from pandemics can be mitigated, is investing in resilient healthcare systems. Such healthcare systems are the cornerstone of limiting the effects of pandemics. A healthcare system can be regarded as resilient when it meets certain criteria, such as being able to treat a sudden surge in patients, related to an unexpected pandemic, or having the appropriate funds to respond accordingly to the emergency. Establishing resilient healthcare systems may face many barriers, such as underpreparedness and the inability to provide basic medical assistance. These barriers need to be mitigated in order to ensure the stability of healthcare systems, as well as the their ability to provide safe care.

In an uneducated world, investing in resilient healthcare systems is both critically necessary and uniquely challenged. Health literacy, which refers to the ability to access, understand and apply medical information, serves as the foundational bridge between healthcare infrastructure and public well-being. When education levels are low, this bridge breaks, making equity the central variable that determines whether a healthcare system survives or collapses during a pandemic. Consequently, equity in an uneducated world significantly affects investment in resilient healthcare systems to mitigate future pandemic risks, as these investments will be based on the needs of each population, which may have limited access to education.

Key Terms and Definitions

Resilient Healthcare Systems

"A resilient healthcare system is one which has the capacity to proactively foresee, absorb, recover from, and adapt to shocks such as pandemics, climate change, geopolitical conflicts, and cyberthreats."[1]

Pandemic

"A pandemic is an outbreak of a disease that occurs over a wide geographic area, such as multiple countries or continents, and typically affects a significant proportion of the population."[2]

Universal Health Coverage (UHC)

"Universal Health Coverage (UHC) means that the entirety of a country's population has access to the full range of essential, quality health services they need, including essential health services, from health promotion to prevention, treatment, rehabilitation and palliative care when and where they need them, without financial hardship."

Pandemic Prevention, Preparedness and Response (Pandemic PPR)

"A global Pandemic Preparedness and Response (PPR) architecture consists of five sub-systems that are interlinked and must be deployed at national, regional and global level: (1) Surveillance, collaborative intelligence, and early warning; (2) Prioritized research and equitable access to countermeasures and essential supplies; (3) Public health and social measures and engaged, resilient communities; (4) Lifesaving, safe and scalable health interventions, and resilient health systems; (5) PPR strategy, coordination, and emergency operations."[3]

Essential/ Primary Healthcare Services

"Primary Health Care is essential health care based on practical, scientifically sound and socially acceptable methods and technology, made universally accessible to individuals and families in the community through their full participation and at a cost that the community and the country can

[1]Organisation for Economic Co-operation and Development. (n.d.). *Health system resilience*. https://www.oecd.org/en/topics/sub-issues/health-system-resilience.html

[2] Merriam-Webster. (n.d.). Pandemic. In *Merriam-Webster.com dictionary*. Retrieved September 30, 2026, from https://www.merriam-webster.com/dictionary/pandemic

[3] People's Health Movement. (2022, March 29). *Analysis of pandemic preparedness and response (PPR) architecture, financing needs, gaps and mechanisms*. WHO Tracker. https://who-track.phmovement.org/analysis-pandemic-preparedness-and-response-ppr-architecture-financing-needs-gaps-and-mechanisms

afford to maintain at every stage of their development in the spirit of self-reliance and self-determination." [4]

Active Pharmaceutical Ingredients (APIs)

"Active Pharmaceutical Ingredients are the main ingredients in a medicine that cause the desired effect of the medicine." They are the ingredients from which medicines are created, and are essential for the production of medicines.

Background Information

The History of Global Pandemics and Healthcare Response

Throughout the years, different pandemic risks have evolved, and along with them, the healthcare response has evolved accordingly. From ancient times to the 19th century, different pandemics have evolved, such as bacterial infections, cholera and smallpox, which primarily spread through trade routes. Most of these pandemics had a great toll on the population at the time, such as the Black Death, which took place from 1347 to 1351, and had an estimated 25 million deaths. Back then, states did not possess a unified healthcare system, resulting in the inability to discuss the their resilience.

In the 19th century, a major scientific shift occurred, which made the transition from remedies and traditional medicine, to modern medicine and scientifically based methods. This occurred thanks to the advances in the understanding and mapping of human anatomy, along with the establishment of physiology as a distinct science. In 1851, the first international sanitary conference took place, where 12 countries convened to discuss the cholera outbreaks occurring at the time. This is one of the first international efforts seen in the healthcare sector, and it is of high significance. Then, in the 20th century, significant scientific leaps occurred, such as the use of vaccines, which are portrayed by the fact that for example, the life expectancy in the UK increased from 48 years, to 72 years of age. In 1948, after the Second World War, the Constitution of the World Health Organization entered into force, a significant landmark in transnational cooperation for healthcare, as this initiated the operations and the global impact and outreach that the World Health Organization has today. Then, in the late 20th century to the 2010s, certain diseases, mostly zoonotic pathogens, such as HIV/AIDS, Ebola and H1N1 were on the watch for the possibility of a pandemic, but major outbreaks occurred, including the 2009 H1N1 influenza pandemic. A resilient healthcare system in such cases will prove beneficial, as it can absorb the sudden surge in patients and their needs, while being able to operate under normal capacity. Most recently, in the 2020s the COVID-19 pandemic was in the spotlight, as the pandemic that put the world to a halt. Hospitals were overflowing, most countries were quarantined, and at the first stages of the disease, before the development of vaccines, the possibility of death was significant. This led to worldwide economic losses, with an estimated 3.4% of the world GDP lost.

[4] University of Cape Town. (n.d.). *What is primary health care?* Primary Health Care Directorate. https://health.uct.ac.za/primary-health-care/phc-approach/what-primary-health-care

Effects & Risks of Pandemics

Overwhelmed Healthcare Systems

Pandemics have a multitude of risks. These stem from systemic shortages, underpreparedness and other factors affecting healthcare systems internationally. Overwhelmed healthcare systems play a major role in the underpreparedness of pandemics. Clinical and ICU surge capacity shortfalls, which practically mean that there are not enough ICU beds, resort to states converting non-clinical areas, such as convention centres or military bases into field hospitals, as seen during the COVID-19 pandemic. Even though these facilities might prove helpful in a crisis, they are not the optimal infrastructure for healing, which might have side effects such as the transmission of illnesses within the medical facility, the degradation of the quality of the care given and increased mortality rates.

Additionally, the shortage of healthcare workers, and the depletion and burnout of the existing doctors is a pressing matter, which stems from the increased number of patients during a pandemic, or the constant, systemic need for doctors in medical facilities. Furthermore, the disruption of essential healthcare services, such as cancellations or delays of surgeries, and the interruption of health programmes, such as childhood vaccination or maternal healthcare pose a risk to the livelihoods of uninfected populations. This can lead, in the long run, to an increased number of hospital visits during pandemics and a spike in secondary excess mortality, referring to deaths caused by delays in emergency interventions, such as heart attacks or strokes, further burdening the overexerted facilities and personnel.

Strained Supply Chains

Strained Supply Chains are also a significant factor of possible mishaps during a pandemic. Active pharmaceutical ingredients (APIs) are vulnerable, due to raw chemical and API manufacturing being concentrated in several countries, including China and India. During a pandemic, transnational trade can be disrupted, affecting the accessibility to said APIs. More specifically, these two states account for a substantial share of global API manufacturing, meaning that if in any case these two states are unable to trade with a state which needs these APIs, there is a high risk that their supply chain will be compromised. Production shutdowns and export restrictions cause severe global shortages in sedatives, antibiotics, pain relief drugs and all medical products that use APIs, which are all the more needed during a pandemic. Export bans, tariffs by exporting nations, reduced global air freight, and port congestion all pose a risk to states which need APIs in order to protect the livelihoods of their citizens.

Disruption of the Economy

Economic disruptions during a pandemic are very significant and often detrimental. As seen during COVID-19, many industries can collapse and states may experience financial crises, along with the medical crisis they are already experiencing. Sharp declines in GDP during pandemics, due to reduced consumer spending, operation halts and border closures, bring

global industries to a halt. Growth in sovereign debt is significant, resulting from emergency stimulus packages, business bailouts and safety net expansions provided by each state during a pandemic crisis. According to the IMF, emerging market governments issued $124 billion in debt during the first six months of 2020, with two-thirds of the borrowing coming in the second quarter. Mass redundancies, furloughs and elevated unemployment rates in service-oriented economic sectors also followed, due to quarantines, during which people are unable to socialise and access services in the tertiary sector. Productivity losses also affect the economy due to widespread illness or long-term disability. Severe revenue losses and insolvencies across high-exposure sectors like travel, aviation, hospitality, retail, and entertainment occur and broad closures among small-medium sized enterprises (SMEs), due to liquidity shortages, deeming them unable to preserve their livelihoods in long periods of inactivity.

Characteristics of Resilient Healthcare Systems

Absorptive Capacity

A characteristic of resilient healthcare systems is absorptive capacity. Absorptive capacity is the ability of a healthcare system to absorb the sudden increase of patients, which happens during a medical crisis. This is done by utilising pre-existing reserves to keep standard operations stable during a crisis, such as a stock of medicine or APIs for medicine production, drawing from strategic national stockpiles of personal protective equipment, activating pre-designed surge beds in hospitals, and deploying trained emergency response teams as an immediate buffer.

Adaptive Capacity

Furthermore, adaptive capacity is key for resilience. A healthcare system can be described as adaptive, when it has characteristics such as flexing operations dynamically when at capacity, offering telemedicine services for routine appointments in order to relieve hospitals, and repurposing general hospital wards into intensive care units. Adaptive capacity is important, as the ability to adapt operations is key for the resilience of public health systems, as it can relieve pressure from hospitals and allow doctors to work from home.

Transformative Capacity

Moreover, transformative capacity is essential in developing efficient and resilient healthcare systems, as well as in pandemic preparedness. Transformative capacity can be achieved through implementing transformative reforms and upgrading the system after a shock, permanently transforming the underlying framework of the system if it proves to be ineffective for sustained security; introducing national health security legislation; establishing special public health emergency authorities; reforming supply chain legislation and reorganising primary care delivery.

Operational Redundancy

Operational Redundancy is the intentional replication of vital supply chain, energy, infrastructure, and production processes in order to ensure no single point of failure exists. It

involves creating alternatives such that the failure of one option does not paralyse the whole system. This is achieved by employing dual sourcing of APIs, establishing local production facilities, provision of power and oxygen in case of an emergency.

The One Health Approach

The One Health approach is an integrated system that acknowledges the close relationship among human, animal, and environmental health. It involves conducting surveillance within all three realms. The One Health approach intervenes with the disease problem at its source in the ecosystem and animals before it jumps to humans. It involves monitoring the wildlife trade and deforestation areas for zoonotic disease spillover, collaborating data exchange between agriculture and health departments, and surveillance of antimicrobial resistance in animals. As such, the One Health approach can reduce the likelihood of a pandemic caused by diseases that originate from wildlife.

The strategy also operates side by side with Pandemic Prevention, Preparedness and Response (PPR) initiatives, whereby the entire lifecycle of a crisis is institutionalised into permanent public planning, governance, and annual budgets, and therefore incorporates a potential crisis into the annual government budget together with the additional capacity provided by hospitals. The strategy aims to shift the current focus of health crisis expenditure from reactive and effect-focused spending to sustained spending and preparedness through the use of multi-agency simulations of outbreaks, continuous genomic surveillance and sustained funding for rapid distribution of vaccines and treatments. This will have a significant impact on future pandemic risks because the planning of a potential health crisis in annual state budgets will lead to more prepared health systems.

Barriers to Resilient Healthcare Systems

Financial Barriers

Financial challenges faced in healthcare systems reflect systematic underfunding, financial instability, and poor management of resources. Less Economically Developed Countries (LEDCs) face these issues much more often than developed countries, since they invest very little of their GDP in healthcare. The ongoing gap makes it impossible for such countries to make some necessary investments in preventive care facilities, reserve fund formation, and hospital upgrading. Even though there may be some trained personnel or facilities available, the lack of working capital makes it impossible to meet even the basic requirements such as consumables and maintenance of equipment.

Institutional Barriers

Institutional barriers arise out of organisational fragmentation, political inertia, and bad governance within the health sector. In many LEDCs and developing health care systems, bureaucratic delays cause immense delays in policy implementation and emergency resource allocation. The disparity in institutional accountability and heavy reliance on centralisation causes delays that render local health

systems paralysed. This organisational friction degrades decision-making efficiency and weakens systemic resilience under prolonged strain.

Workforce Barriers

Workforce barriers include shortages, maldistribution, and burnout of medical practitioners. The problem of brain drain is always prevalent in LEDCs, where highly trained nurses and doctors move to other areas with better-paying jobs, causing understaffing in their local hospitals. In case of any prolonged health shocks or disease outbreak, the current staff will be subjected to extreme physical tiredness, mental pressure, and even unsafe working conditions, thus increasing absenteeism and turnover rates.

Supply Chain Constraints

Supply chain vulnerabilities derive from market dependencies, logistical bottlenecks, and inadequate manufacturing capacity. Modern health institutions tend to depend greatly on global trade networks for the supply of APIs, medical products, and equipment such as ventilators. During international emergencies, limitations in trade and panic buying put LEDCs at the bottom of the supply chain, leading to shortages within the region. Health institutions will be unable to operate normally due to lack of regional manufacturing capacity and sufficient regional buffers.[5]

Access Barriers

Financial barriers arise in LEDCs because of out-of-pocket costs as well as the non-existence of universal health insurance coverage. Social prejudice, gender discrimination, and stigmatization are common barriers which often hinder women and other disadvantaged groups from seeking care on their own. Physical barriers such as lack of proper roads, inadequate public transport, and regional clinics force people living in rural areas to travel long distances in order to access care. These all impact pandemic preparedness and the efficiency of healthcare systems, as these issues which healthcare systems face daily can worsen operational capacity. [6]

Operational Barriers

Operational barriers include the lack of physical infrastructure and technology that affect the provision of care on a daily basis. The most important example is the lack of ICU beds and advanced monitoring devices, which rapidly results in system breakdown when there is an increase in the number of patients. Moreover, many hospitals fail to have the necessary Early Warning Systems

[5] World Health Organization. (2021). *Building health systems resilience for universal health coverage and health security during the COVID-19 pandemic and beyond: WHO position paper*. World Health Organization.

[6] Terefe, B., Workneh, B. S., Zeleke, G. A., Mekonen, E. G., Zegeye, A. F., Aemro, A., Tekeba, B., Tamir, T. T., Wassie, M., & Ali, M. S. (2025). Uncovering women's healthcare access challenges in low- and middle-income countries using mixed effects modelling approach: Insights for achieving the Sustainable Development Goals. *PLOS ONE*, *20*(1), e0314309. https://doi.org/10.1371/journal.pone.0314309

(EWS) and digital health monitoring tools that can help identify the deterioration of clinical conditions or epidemics at an early stage.[7]

Timeline of Events

Date	Description of Event
23 July 1851	The 1st International Sanitary Conference takes place, where twelve nations convened in Paris following the devastating Cholera epidemics across Europe. [8]
7 April 1948	The WHO Constitution enters into force, which created the world's primary directing and coordinating authority on international public health. [9]
25 July 1953	The UN Treaty series registers the constitution of the WHO ; it had already entered into force in 1948.
25 July 1969	Adoption of the International Health Regulations, which revised the 1951 international sanitary regulations. [10]
26 April 2000	The WHO officially launches the Global Outbreak Alert and Response network.[11]
12 March 2003	The WHO issues a global alert on SARS, a novelty respiratory virus which rapidly spread across Asia and North America.[12]

[7] World Health Organization. (2021). *Building health systems resilience for universal health coverage and health security during the COVID-19 pandemic and beyond: WHO position paper*. World Health Organization.

[8] The Clinical Times. (n.d.). *1851: The First International Sanitary Conference*. https://theclinicaltimes.com/landmarks/1851-the-first-international-sanitary-conference

[9] World Health Organization. (n.d.-c). *Constitution*. https://www.who.int/about/governance/constitution

[10] United Nations. (1953). *Constitution of the World Health Organization* (United Nations Treaty Series, Vol. 175, No. I-2303, pp. 185–211). https://treaties.un.org/doc/Publication/UNTS/Volume%20175/volume-175-I-2303-English.pdf

[11] World Health Organization. (n.d.-d). *The Global Outbreak Alert and Response Network (GOARN)*. World Health Organization Regional Office for the Western Pacific. https://www.who.int/westernpacific/wpro-emergencies/response/the-global-outbreak-alert-and-response-network-(goarn

[12] World Health Organization. (2003, April 11). *Severe acute respiratory syndrome (SARS) - multi-country outbreak - Update 27*. https://www.who.int/emergencies/disease-outbreak-news/item/2003_04_11-en

11 March 2020	The WHO characterises COVID-19 as a pandemic, after SARS-Cov-2 spread to more than 110 countries within weeks.
1 June 2024	The WHO adopts modernized International Health Regulation IHR amendments, at the 77th world health assembly, introducing alert mechanisms and mechanisms updated to enforce financing and equitable access to obligations.

Major Actors Involved

United States of America

The United States of America is among one of the largest financial contributors for international health programmes. The United States of America is financing biomedical research and disease surveillance globally. They have also assisted the creation of response programmes, to minimise future pandemic risks, and build more resilient healthcare systems. With its agencies, such as the Centers for Disease Control and Prevention (CDC), they contribute substantially to the prevention and control of infectious diseases. The United States is also one of the global leaders in the production of pharmaceutical products, such as medication or protective devices, making it a key supplier of the North American and European medical market.

People's Republic of China

China, as reflected in Figure 1, is a major producer of active pharmaceutical ingredients, which are supplied worldwide, and are essential for the production of even the most basic vaccines and medicine. Additionally, they are a strategic supplier for medicine and pharmaceutical products, as well as personal protection devices. The PRC demonstrated rapid emergency infrastructure mobilisation during the COVID-19 pandemic, although the overall response also faced serious challenges. China built makeshift hospitals within days, went on lockdown and implemented extensive measures intended to protect public health.

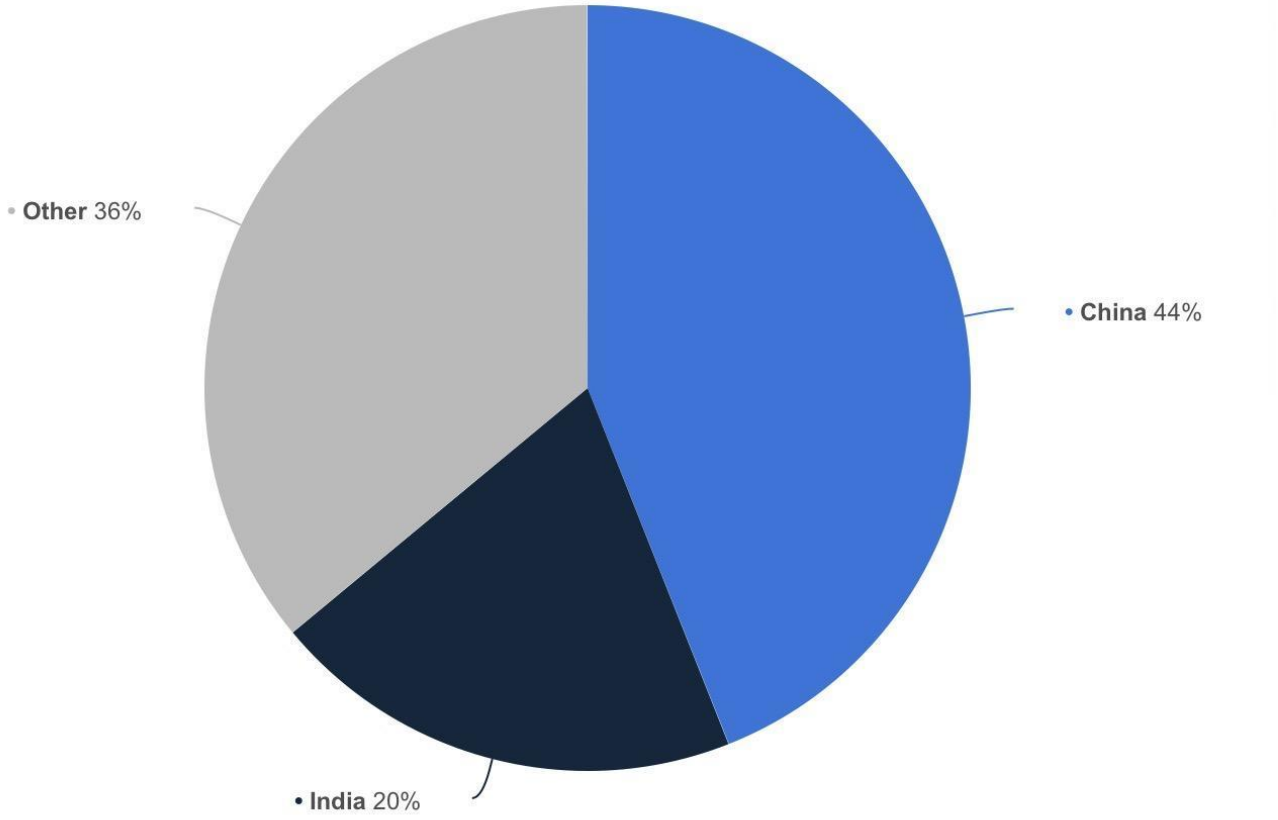


Figure 1: Distribution of active pharmaceutical ingredient (API) production volume worldwide by region as of 2023[13]

Republic of India

The Republic of India is known internationally as the 'pharmacy of the world'. India is a nation with an extensive capability in generic drugs manufacturing and vaccines production, along with API production. As seen in figure 1, India manufactures a substantial share of the world's APIs, only behind China. India provides assistance to Less Economically Developed Countries (LEDCs) at affordable prices, with initiatives such as the Vaccine Maitri, which supplied hundreds of millions of COVID-19 vaccine doses internationally, delivering significant quantities as grant assistance or via affordable bilateral agreements to LEDCs across Africa, South Asia, and the Caribbean. However during health crises, the country experiences pressure on health supplies and infrastructure, due to its population, which exceeds 1 billion people. This can place the Indian healthcare system under severe strain during a sudden influx of patients, which will be significant due to the sheer population of the country.

[13] Statista. (n.d.). *API production volume by region*. https://www.statista.com/statistics/1498065/api-production-volume-by-region/

Federative Republic of Brazil

The Federative Republic of Brazil is an important regional player in the Latin American pharmaceutical industry, where domestic drug producers account for more than 50 percent of the total retail turnover in pharmaceuticals of the country. Featuring important domestic capabilities in terms of vaccine and generic drugs production, Brazil features important domestic production facilities like Bio-Manguinhos and Instituto Butantan. The country plays a very important role in providing assistance to the neighbouring Less Economically Developed Countries (LEDCs) in Latin America and Lusophone regions through South-South Cooperation, offering affordable access to essential generic drugs and antiretrovirals. However, in case of important public health challenges, Brazil experiences extreme strain on its national unified health care system (SUS) and domestic supply chains, due to the country's population that exceeds 200 million people.

Gavi, the Vaccine Alliance

Gavi, the Vaccine Alliance, is a multilateral public-private partnership that brings together global purchasing power to improve vaccine equity and boost immunisation rates in developing countries. Via its novel market shaping and Advance Market Commitment (AMC) funding mechanisms, Gavi motivates pharmaceutical companies to reduce costs and increase production volumes for Less Economically Developed Countries (LEDCs). Over its existence, Gavi has supported the immunisation of more than 1 billion children, saved millions of lives in the future and established critical cold chain facilities in the member countries. But in times of a global public health emergency, Gavi is constrained in its operations because of its reliance on voluntary donations, capital constraints, and lack of regulatory authority to address vaccine hoarding by developed nations. Thus, while Gavi’s co-funding approach helps build sustainability in normal circumstances, it is necessary to link its funding structure to emergency capital funds and decentralised manufacturing facilities to ensure crisis resilience.

World Health Organization (WHO)

The World Health Organization (WHO) is essential in reducing the risk of future pandemics through investment in robust health systems. The WHO supports nations in setting up early warning systems and biological monitoring through instruments such as the International Health Regulations (IHR) and the Global Outbreak Alert and Response Network (GOARN). In an attempt to surmount the extreme challenges in operations during health emergencies, the WHO offers technical assistance, dispatches Emergency Medical Teams to bolster inadequate ICU structures, and manages global logistics hubs for securing critical pharmaceutical supply chains. Moreover, the WHO works towards addressing the financial, staffing, and access challenges facing developing countries through

universal health coverage financing, global standards to counteract medical brain drain, and gender-sensitive access to essential health care services.[14]

World Bank

The World Bank, an international financial institution that provides long-term development loans and grants to low- and middle-income nations, plays a critical part in the financing development of global health security. Besides offering financial resources, assistance is frequently needed. Therefore, in the development of national health care systems, the World Bank acts as a trustee and host of the Pandemic Fund Secretariat. This independently governed financial facility, established with WHO involvement, offers longer-term resources, to further enhance the preparation for pandemics, and early warning systems, showcasing that the World Bank can act as a contributor in the investment in resilient healthcare systems.

UN Documents and International Agreements

United Nations General Assembly (UNGA) A/RES/67/81

UN General Assembly resolution 67/81 entitled "Global health and foreign policy," adopted on 12 December 2012, was a major shift through the incorporation of public health into the agenda of global development and foreign policy. The resolution recognises that health not only affects but is affected by sustainable development and therefore urges for the establishment of strong multilateral cooperation and commitment at the national level to achieve Universal Health Coverage (UHC). Through its focus on providing access to high-quality and affordable health care services without financial barriers, A/RES/67/81 highlights the main goal of the United Nations, which cooperates closely with other international organizations like the World Bank to establish sustainable health systems in the countries. However, being a non-legally binding document, it does not have any enforcement or funding mechanisms to ensure the national budget allocation for the matter. Thus, even though it helped to highlight UHC as a main principle of SDG 3.8, its influence is limited by the financial restrictions, political resistance, and health inequalities in LEDCs.

World Health Assembly (WHA) Resolution WHA77.17

Passed by the World Health Assembly on 1 June 2024, Resolution WHA77.17 revises the 2005 International Health Regulations (IHR) to enhance global pandemic preparedness. The revised IHR includes "Pandemic Emergency" as the highest alert category and provides for National IHR Authorities and a Coordinating Financial Mechanism to support implementation in Less

[14] World Health Organization. (n.d.). *Our work*. https://www.who.int/our-work

Economically Developed Countries (LEDCs). Through the coupling of surveillance and financing, Resolution WHA77.17 updates global health security. Its effects are hindered by limited direct enforcement provisions or automatic funding agreements from wealthier countries. Nevertheless, the effect of this resolution is in the development of clearly defined criteria, resource equity as an international health law issue, and the provision of a structure for funding in low-resource areas. In the end, this system changes global emergency management from reactive and nationalist border controls to a framework for improved cooperation and financing, without guaranteeing automatic funding.

WHO Pandemic Agreement (WHA78.1)

Whereas Resolution WHA78.1 is a groundbreaking achievement in the codification of global health solidarity, the future effectiveness of the WHO Pandemic Agreement will largely depend on its ability to overcome enforcement shortcomings, intellectual property disputes, and long-term financing problems. While the proposed Pathogen Access and Benefit-Sharing (PABS) System and the One Health approach are aimed directly at addressing the immediate problems identified in the context of the COVID-19 pandemic, the explicit recognition of the principle of national sovereignty means that implementation will depend on states' ratification and compliance, without having any effective means of enforcing penalties for non-compliance or nationalistic export policies in case of an ongoing crisis. In addition, although the envisaged financing mechanisms could facilitate resource transfers, their dependency on continuous political goodwill instead of obligatory funding makes it vulnerable to leaving health systems in LMICs underfinanced between epidemics. Therefore, the adopted text seeks to update international health law from reactive to structural, but its success will depend on whether the member states live up to their commitments equitably despite their national interests in the next global health crisis.

Previous Attempts to Solve the issue

The COVAX Facility

The COVAX Facility, led by Gavi the Vaccine Alliance, CEPI, the World Health Organization (WHO), and UNICEF, served as the global procurement mechanism designed to counter vaccine nationalism during COVID-19. The main significance of COVAX in this regard was its ability to operationalize equitable access through pooling financial resources for the procurement and distribution of vaccines to LEDCs. However, COVAX had several weaknesses that affected its efficacy. It lacked sufficient early financing and manufacturing commitments, as well as regulatory authority that could have prevented wealthy countries from securing early supply of vaccines through bilateral agreements. In addition, COVAX struggled with late deliveries due to export bans in the main manufacturing

centers, inadequate cold-chain capacity, and erratic funding. An analysis of COVAX indicates a complicated legacy. At the end of COVAX operations, nearly 2 billion vaccine doses had been delivered, with estimates attributing millions of lives saved to vaccination programmes supported by COVAX. This shows that multilateral procurement can work effectively even when undertaken on such a massive scale. Nevertheless, the initial difficulties experienced by COVAX show that using voluntary market forces in a situation of an ongoing health crisis is not effective in providing immediate equity.

The Global Health Security Agenda (GHSA)

Launched in 2014, the Global Health Security Agenda (GHSA) is an international initiative that brings together more than 70 countries and organizations in order to improve adherence to the WHO International Health Regulations. The main role of GHSA in addressing pandemic risks is the development of surveillance and laboratory capacities in Less Economically Developed Countries (LEDCs). Applying a technocentric approach, GHSA supplied crucial technological equipment and digital surveillance systems, which helped to set up the baseline of disease surveillance in LEDCs. Nevertheless, there were some important operational drawbacks associated with the purely technocentric approach of GHSA in managing population risks. Excessive dependence on technology was hampered by poor digital infrastructure, power instability, and the lack of properly trained personnel for using complex devices. Moreover, advanced surveillance cannot stop outbreaks without investing in primary care and logistics simultaneously. Improving the evaluation reveals that while technical capacity is necessary for early detection, it is not enough to create a sustainable healthcare system. GHSA demonstrates that digital tools and laboratory upgrades are important elements of pandemic preparedness; however, health security needs more than just technical infrastructure.[15]

The World Bank Pandemic Fund

Established by the World Bank Group in late 2022, the Pandemic Fund represents a dedicated financial mechanism designed to help bridge systemic gaps in global pandemic prevention, preparedness, and response (PPR) across low- and middle-income countries. Addressing historic vulnerabilities where LEDCs lacked immediate access to capital during health crises, the fund directs targeted grants toward strengthening early-warning disease surveillance networks, expanding regional laboratory capacities, and training frontline health workforces. By operating as a collaborative platform that co-finances projects alongside national governments, regional entities, and international public health institutions, the fund aims to build resilient, adaptable health

[15] Global Health Security Agenda. (n.d.). *Global Health Security Agenda*. https://globalhealthsecurityagenda.org/

systems capable of detecting and containing localized biological threats before they escalate into global emergencies.

Possible Solutions

Global Price Transparency Initiatives for APIs / critical medication

Transparency in the pricing of Active Pharmaceutical Ingredients (APIs) and final critical medications addresses issues of supply chain exploitation, market manipulation, and price gouging in both normal times and in the case of health emergencies. Public disclosure of the net cost of transactions, production cost, and research expenses as encouraged by World Health Assembly Resolution WHA72.8 could enable member countries to resolve the issue of asymmetric information between governments and multinational producers. Transparent cost structures facilitate the use of international reference pricing, cost comparisons across borders, and multicountry pooled purchasing agreements. This can increase the leverage of the Less Economically Developed Countries (LEDCs) in negotiations. Ultimately, lower costs of acquisition of vital active ingredients can help reduce shortages and monopolistic practices by corporations.

Improved Workforce Retention and Expansion Initiatives

Maintaining the resilience of the health system entails expanding human resources through specific strategies for retention, capacity building, and distribution. The LEDCs suffer from a significant "brain drain," a high attrition rate, and distribution problems in rural areas, which make their health systems extremely fragile during epidemic situations. Overcoming these weaknesses requires task-shifting strategies, professional accreditation in regions, competitive remuneration packages, and continuous training. With the help of multilateral financing mechanisms like World Bank loans and technical assistance from the World Health Organization, expansion and retention of the health workforce will allow uninterrupted delivery of routine services and emergency readiness at the same time. A protected and well-paid health workforce is necessary to turn medical supplies, surveillance information, and emergency plans into practice.

Creation of an Oversight Mechanism

The creation of an independent multilateral oversight committee could improve accountability, compliance and efficiency in terms of financial and resource disbursement within global health initiatives. Through its coordination with National IHR Authorities, the committee assesses regulatory harmonization, identifies emergency supply chain challenges, conducts audits of aid disbursements, and follows up on non-binding policy declarations. Through the examination of financial transactions from organizations such as the World Bank's Pandemic Fund, the process can reduce opportunities for corruption, misuse and inefficiency in handling crises. The audits and reporting mechanisms ensure that there is mutual accountability between donor countries and

receiving countries, thus ensuring that emergency funds, technical aid, and medical supplies are allocated appropriately based on objective epidemiological need.

Establishing independent local manufacturing networks

Development of local manufacturing networks via intergovernmental developmental organizations is an important approach to enhancing resilience in healthcare systems through the diversification of manufacturing capabilities for medical supplies, diagnostic machinery, and drugs. Traditionally, countries have been at risk of facing shortages, export bans, and increased prices for critical health products due to their dependency on centralized global supply chains during global health crises. Regional manufacturing capabilities can be developed through technology transfers, flexible financing instruments, and regional regulatory harmonization in order to achieve more reliable access to critical health products suited for each country's population. Moreover, regular production within the country allows keeping the necessary infrastructure, workforce, and supply chains functional in times of peace, which will make it possible to quickly ramp up production in case of a crisis.

Investing in early warning systems

Investment in early warning systems transforms the approach to global health security from being reactive in nature to being proactive against potential threats. Through investment in pandemic surveillance systems, via the World Bank or any other financial institution, it is possible to establish measures such as community symptom surveillance, laboratory diagnostics, and environmental surveillance. The countries will be able to identify any biological threats at an early stage, before they become widespread. If such systems are run alongside the current public health infrastructure and through automated data transmission mechanisms, then such systems will give the state's health ministers sufficient time to activate the necessary protocols for containment, allocate emergency funding, and distribute the necessary medical supplies. In essence, early warning systems become the first line of defence for national health systems, reducing the burden that epidemics place on them.

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